
PS Analysis in Data Science: Problem Set 9

Problem 1. Consider $A \in \mathbb{R}^{m \times n}$ and prove the following claims made in class:

- (a) If $\mathbb{R}^m$ and $\mathbb{R}^n$ are equipped with the sum norm, then the operator norm of A is the column sum norm $\|A\|_1$.
- (b) If $\mathbb{R}^m$ and $\mathbb{R}^n$ are equipped with the max norm, then the operator norm of A is the row sum norm $\|A\|_\infty$.
- (c) Find matrices A such that
 - $\|A\|_1 = \|A\|_\infty$,
 - $\|A\|_1 > \|A\|_\infty$,
 - $\|A\|_1 < \|A\|_\infty$.

Problem 2. Connections between geometric sum and von Neumann series:

- (a) Implement the geometric sum $\sum_{k=0}^n x^k$ in Julia as a function *myGeometricSum*(x, n) by simply calling the Julia function *myNeumann* from the lecture notes. Do you see the connections?
- (b) Let B be a Banach space. Prove by induction over n : if $g \in \mathcal{L}(B)$ is such that $\text{id}_B - g$ is invertible in $\mathcal{L}(B)$, then

$$\sum_{k=0}^n g^k = (\text{id}_B - g^{n+1})(\text{id}_B - g)^{-1}.$$

Problem 3. Consider the matrix

$$A = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$$

- (a) Determine the inverse of A by computing the associated von Neumann series by hand.
- (b) Implement the von Neumann iteration and compute the inverse of B in Julia:

$$B = \begin{pmatrix} 1.20 & 0.05 & 0 & 0 & 0.02 \\ 0.03 & 0.90 & 0.04 & 0 & 0 \\ 0 & 0.02 & 1.10 & -0.03 & 0 \\ 0 & 0 & 0.05 & 0.85 & 0.04 \\ 0.01 & 0 & 0 & 0.02 & 1.15 \end{pmatrix}$$

- (c) Use “Rational{BigInt}” in Julia and check that “inv(B)” and the von Neumann series computed till the first $n = 150$ terms in Rational{BigInt} deviate by at most 10^{-100} in each entry.

Problem 4. Consider the space

$$\mathcal{C}([0, 1]) := \{f : [0, 1] \rightarrow \mathbb{R} : f \text{ continuous}\}.$$

(a) Prove that $\mathcal{C}([0, 1])$ is a normed, linear vector space with norm

$$\|f\| := \int_0^1 |f(x)| dx.$$

(You may assume that this is a norm. . .)

(b) Show that the map $S : \mathcal{C}([0, 1]) \rightarrow \mathbb{R}$,

$$S(f) = f(0)$$

is linear but not continuous.

(c) Is there an alternative norm on $\mathcal{C}([0, 1])$ such that S is continuous?